

Lesson 10

- 3 As shown in the picture, there are three cards, each with a number on it. If we pick one or more cards from them (we may not rotate the cards) to form some numbers, we can get _____ different prime numbers.



- A. 3 B. 4 C. 5 D. 6 E. 7

L10 Extention

- 7 Given n is a certain natural number in $1 \sim 2021$. If the 72 multiple of n is a perfect square, then the maximum value of n is _____ .
- 8 There are some two-digit numbers, each of which is still a prime number after swapping the first and the second digit. The sum of all these numbers is _____ .

- 9 If we use integers from 1 ~ 9 to form several prime numbers while each integer can be used only once, we can form at most _____ prime numbers.

- 2 Find the number of factors of 196, which is _____ .

- 9 The number of natural numbers with exactly 10 factors less than 100 is _____ .

Lesson 11

2 Fill in the blank.

(1) $(5250)_6 = (\quad)_{10}$.

(2) $(2564)_{10} = (\quad)_8$.

6 The remainder of a three-digit number is always 2 when it is divided by 3, 5, or 7. What is the sum of digit of the largest value of such a number? _____.

A. 20

B. 18

C. 16

D. 14

E. 12

L11 extension

7 What is the value of a two-digit number which is 9 times the sum of its digits?

- 6 Dividing a certain two-digit number by 10 leaves a remainder of 9, and dividing it by 9 leaves a remainder of 8. The sum of the number's two digits is _____ .

A. 7 B. 9 C. 13 D. 17

- 8 After adding 72 to a 2-digit number, the tens digit and the ones digit are reversed. What is the sum of all the possible 2-digit numbers? _____ .

- 7 The sum of the following numbers in base-10 is _____ .

$(214)_5$ $(ABA)_{16}$ $(1234)_8$

- 9 After adding 54 to a 2-digit number, the tens digit and the ones digit are reversed. What is the sum of all the possible 2-digit numbers? _____ .

Lesson 12

4 Simplify the following expressions:

(1) $-\frac{2a-4}{2-a} = \underline{\hspace{2cm}}$
A. -2

B. 2

8 Simplify:

(2) $\frac{a^2 - b^2}{a^2 - 2ab + b^2} \div \frac{2a - 2b}{a + b} = \underline{\hspace{2cm}} .$

L12 extension

(2) $\frac{-y^2}{3x^2y} \div (3xy^2) = \underline{\hspace{2cm}}$

A. $-\frac{1}{9x^3y}$

B. $\frac{1}{9x^3y}$

C. $-\frac{y^2}{9x^3}$

D. $-\frac{1}{x^3y}$

$$(2) \frac{4m^2 - 8m + 4}{m^2 - m} = \underline{\hspace{2cm}}$$

A. $\frac{(m-1)}{m}$

B. $\frac{4}{m}$

C. $\frac{4(m+1)}{m}$

D. $\frac{4(m-1)}{m}$

7 Simplify:

$$(1) \frac{2x}{x-3} \div \frac{x^2}{2x-6} = \underline{\hspace{2cm}}$$